Deepfake Detection Using Fusion-Based Deep Learning: A Reproducibility Study of DeepfakeCatcher

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***Abstract*—The recent improvement of generative AI has made the creation of almost realistic deepfakes a big threat to digital trust and cybersecurity. This paper shows a study of the *Deepfake Catcher* framework that was reproduced by us, Making sure whether a fusion based deep learning model can effectively outperform complicated architectures. We implement a dual backbone system using EfficientNet-B4 and Xception with feature level fusion,We trained it on a subset of the FaceForensics++ dataset. Our results showed a successful replication, we achieved a 97.49% AUC-ROC and 91.00% Accuracy, within a 2% margin of the original published paper results. Although we faced moderate overfitting due to dataset size constraints, the results findings can confirm that the combined result of spatial and texture features provides a robust and resource efficient solution in the era of digital forgery.**

***Index Terms*—deepfake detection, feature level fusion, CNN backbones, Reproducibility study, FaceForensics++(dataset), EfficientNet, Xception, transfer learning**

I. Introduction

A. Research Problem

The recent rapid advancement of AI and generative techniques has made the creation of highly realistic manipulated deepfake images and videos increasingly accessible. These manipulative media pose serious threats to politics, cybersecurity, identity protection, and digital trust. Traditional forensic methods that relied on handcrafted features have failed to detect subtle artifacts in high-definition images and videos.

The central research problem is therefore: *"How can supervised deep representation learning help to develop generalizable models that accurately detect deepfakes across diverse manipulation techniques, while maintaining scalability and resilience to unseen generation methods?"*

B. Motivation

The rapid improvement of generative AI has made the creation of highly realistic deepfakes trivially accessible. This poses severe threats to society, including political misinformation, false news, identity theft, financial fraud, and non-consensual synthetic content generation. Developing accurate deepfake detection tools is essential to preserving media trust, protecting user privacy, and enabling the removal of harmful synthetic content.

As Goodfellow et al. [2] said in their discussion on the "curse of dimensionality," the traditional machine learning algorithms we use struggled to generalize when they are faced with data that is very complex,has high dimensional inputs. Deep learning is very much suited for this scenario because it uses learned representations: instead of using human selected features, a deep neural network can automatically discover the faint, complex patterns that are left behind by deepfake generation algorithms.

The goal of this work is to able to identify deepfakes at both high and low resolution from real world internet content, therefore restoring confidence in digital media without relying on community.

C. Overview of the Selected Paper

The paper under study, *Deepfake Catcher: Can a Simple Fusion be Effective and Outperform Complex DNNs?* [1] (Agarwal & Ratha, CVPR 2024), proposes that each pretrained CNN should be divided at a "break point" into three components:

- **Fixed Knowledge (Part A):** Early layers are frozen, retaining generic, architecture-independent features such as edges and textures.
- **Adaptive Knowledge (Part B):** Middle layers are fine-tuned for the deepfake detection task, adjusting high-level representations.
- **New Knowledge (Part C):** Randomly initialized layers appended to the backbone, entirely task-specific and unbiased by ImageNet pretraining.

Multiple CNN backbones (VGG-16, MobileNet, Xception) are each trained independently with this structure, and their softmax outputs are combined using a weighted score fusion. The paper reports 98.01% AUC on FF++ HQ, 97.01% on FF++ LQ, 91.26% accuracy on the Deepfakes dataset, and 0.7035 AUC on cross-dataset Celeb-DF2.

II. Background

The paper is built on the observation that deep CNNs pretrained on ImageNet tend to learn similar low-level features in their early layers regardless of architecture. Key related approaches in the field include:

A. CNN-Based Spatial Artifact Detection

Convolutional Neural Networks (CNNs) are the foundation for deepfake detection due to their ability to extract subtle hierarchical features. Models like Xception or EfficientNet are trained to identify "low-level" traces that can be left behind by image generation processes:like blurred edges around the persons' face, mismatching lights between the face and background or unnatural texture [10].

The primary bottleneck or Limitation is the small temporal scope. Because in these models we treat a video as not a video but a collections of frames in that video. Since we have static frames therefore we cannot catch the "Flickering" or "jitter" that appears when a video is running continuously. As the Generative Adverserial Network become more and more complicated at producing high frequency textures, spatial only detectors will require more complex preprocessing like Laplacian of Gaussian (LoG); a preproceesing technique in which we reduce noise and apply Laplacian operators so that we can see which areas has rapidly changed.

B. Temporal-Based Detection Methods

Temporal methods have changed focus from *what* a face looks like to *how* it moves. Architectures like Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) and CNNs analyze the coherence of a frames over time [12]. They are very effective at spotting irregularities that can be difficult for AI to reproduce like irregular eye blinking motion, unnatural lip syncing of the person in video.

While they are very effective against poorly rendered videos, these methods are computationally very expensive. Processing 30–60 frames per second as a 3D volume requires high and significant VRAM, this creates a bottleneck for real time deployment of model.

C. Transformer-Based Approaches

Following the attention paradigm [13], Vision Transformers (ViTs) use self-attention mechanisms to relate distant parts of a given image. For example, a Transformer can analyze symmetry between the left and right eye, or consistency between the chin and neck line. Models such as the Cross-Attention Transformer allow the network to learn which specific image patches are most likely to contain AI generation evidence. However, this global perspective comes at a high computational cost: Transformers require large data to converge and possess quadratic complexity relative to the number of image patches.

D. Dataset-Driven Benchmark Methods

Benchmark focused research such as the **DeepFake Detection Challenge (DFDC)** [5] and **FaceForensics++** [3] provides standardized "proving grounds" for new algorithms. These datasets are carefully selected to include various compression levels, lighting conditions and diverse demographics, making sure that models are not only memorizing a specific type of artifact. By providing standardized train/val/test splits, they allow the global research community to compare spatial, temporal, and transformer models on a level playing field.

E. Generalization-Based Methods

Cross dataset generalization is one of the challenges in deepfake detection. Most models are good at detecting the specific manipulation type they were trained on (e.g., FaceSwap) but fail when unseen methods like e.g., StarGAN. Generalization methods often use high frequency features or noise patterns that are common to all GANs regardless of their architecture [11]. Now instead of us teaching the AI to recognize specific known forgery techniques .We train it to identify the distinctive mathematical patterns of forged generation. The only trade off here is that these models are less accurate on specific datasets and they require advance training pipelines.

III. Model Description and Proposed Architecture

A. Implemented Model Overview

While the original paper uses the score level fusion of multiple CNN backbones (VGG-16, MobileNet, Xception) .This new implementation uses a **feature level fusion** approach where it uses two pretrained architectures a) EfficientNet-B4 [6] and b)Xception [7]. Each network acts as an independent feature extractor and their learned features are concatenated right before classification.

B. Architectural Details

1) Branch Design: Each branch follows a transfer learning setup.It is pretrained on ImageNet, with final classification layers removed so it gives high level feature embeddings output.

TABLE I
Backbone Branch Output Dimensions

Branch	Backbone	Output Feature Size
1	EfficientNet-B4	1792
2	Xception	2048

2) Feature Fusion: Feature vectors from both branches are appended which then results in a fused vector of $1792 + 2048 = 3840$ total dimensions.

3) Classification Head: The fused representation is then passed through a fully connected classifier.The following is the flow.

$$\text{Input}(3840) \rightarrow \text{FC}(1024) \rightarrow \text{BN} \rightarrow \text{ReLU} \rightarrow \text{Drop}(0.5)$$

$$\rightarrow \text{FC}(256) \rightarrow \text{ReLU} \rightarrow \text{Drop}(0.25) \rightarrow \text{Output}(2) \rightarrow \text{Softmax}$$

producing a binary Real/Fake classification.

C. Motivation Behind Design Choices

1) Feature Level vs. Score Level Fusion: **Score level fusion** (late fusion) is a “shallow feature fusion” approach where each of the model does the work independently and after that their final probability outputs are averaged or weighted. This can miss complicated cross model relationships.

Feature-level fusion (early/mid-fusion), used in this work, concatenates high dimensional feature vectors from different backbones before the classification stage. This preserves a richer, multi-dimensional representation and allows the network to learn cross domain correlations—for example, how a spatial artifact captured by Xception reacts to a texture irreguarity captured by EfficientNet—leading to a more better and discriminative decision boundary.

2) Choice of Architectures: **EfficientNet-B4** applies compound scaling so that it balance network depth of frames, width, and resolution, thus providing a good trade off between parameters efficiency and their accuracy. It also adept at capturing fine grained spatial features that are often times smoothed out by shallower networks.

Xception relies on depthwise separable convolutions.It processes spatial correlations and cross channel correlations independently. In deepfake detection this is important for identifying the micro “checkerboard” artifacts that are commonly left behind by GAN based upsampling. Combining both introduces a balancing feature learning.

3) Transfer Learning Strategy: Training a deep neural network from a state of random initialization requires massive datasets and large computational power. By utilizing Transfer Learning, we assign weights pre trained on the ImageNet dataset. These weights already contain the previous knowledge of how to identify basic shapes, edges, and texture features that are universal across all computer vision tasks. In our strategy, the base convolutional layers are frozen, while only the classifier layers are trained from scratch (because features were extracted independently) .This allows the model to fine tune its advanced reasoning to the task of forgery detection while retaining the robust feature extraction capabilities learned from millions of natural images. This approach significantly lowers the training time for the model, reducing training time and preventing the model from failing on limited deepfake samples.

4) Regularization Techniques: To make sure the model performs as well on unseen internet videos as it does on our training set, we employ three key regularization techniques:

Dropout (0.5, 0.25): Dropout acts as a ”regularizer” by randomly deactivating a percentage of neurons during each training pass. This helps to prevent the network from becoming overly dependent on a specific path or memorizing the training data . It forces the remaining neurons to learn more redundant and robust features because this forces them to learn , effectively creating an internal group of sub networks.

Batch Normalization: Deep networks often suffer from Internal Covariate Shift, where the distribution of inputs to each layer changes during training. Batch Normalization rescales and offsets the data within each mini batch. This stabilizes the learning process and allows for higher learning rates, it also acts as a regularizer, significantly accelerating convergence.

Early Stopping: This is a technique against overfitting. By monitoring the loss on a separate validation set, the training process is stopped the moment the model begins to learn the noise rather than the actual patterns. This ensures the final model captures the general characteristics of a deepfake rather than the specific learning specific noises of a particular training file.

IV. Experimental Setup

A. Hardware and Software Environment

TABLE II
Hardware and Software Comparison

Feature	Our Implementation	Original Paper
Hardware	Google Colab T4 (16 GB)	NVIDIA RTX 2080
Framework	PyTorch 2.x	Keras / TensorFlow
Dataset Size	∼4,000 frames	Full FF++
Training Epochs	13 (early stopping)	50
Training Time	∼25 minutes	Not specified

B. Dataset

The FaceForensics++ dataset was used in our experiment, which is a standard benchmark for deepfake detection models. It contains manipulated videos that are generated using methods like Deep Fakes, Face 2 Face, Face Swap, Neural Textures and Face Shifter, In both high and low quality formats so we can test robustness under compression. Since the full dataset is very large (about 200 GB), a smaller Kaggle subset (hungle3401/faceforensics, which is 2.73 GB) was used in our experiment. It has around 200 real and 200 fake videos making sure that the dataset is balanced for fair model training.

C. Preprocessing

Frames were extracted using library OpenCV, From every video 10 frames were extracted, totaling in about 4,000 frames. Unlike the original paper, face detection was not applied due to hardware bottleneck. To help the model for better generalization, We applied a few standard training augmentations like RandomHorizontalFlip and ColorJitter. However, we kept the validation and test sets ’clean’ meaning only resizing and normalizing them, so that we can ensure we were evaluating the model on consistent, real world data.

D. Training Configuration

The Adam optimizer [14] was used with a weight decay of 1e-4. A ReduceLROnPlateau scheduler lowered the learning rate when validation loss stopped improving. Early stopping with patience=5 was applied so that we can prevent overfitting.

TABLE III
FULL SETUP COMPARISON

Component	Paper Setup	Our Setup
GPU	RTX 2080	Colab T4
Framework	Keras/TF	PyTorch 2.x
Backbones	VGG-16, MobileNet, Xception	EfficientNet-B4, Xception
Fusion Type	Score-level	Feature-level
Dataset	Full FF++ (∼1000 videos)	Kaggle (∼400 videos)
Face Detection	Viola-Jones cropping	None
Epochs	50	13 (early stopping)

V. RESULTS

A. Evaluation Metrics

Accuracy Measures the proportion of correctly classified samples across both the classes.Since we had a balanced FF++ subset i.e (50% real videos, 50% fake) videos, accuracy was a reliable metric.

AUC-ROC A threshold that is an independent metric evaluating how well the model separates the two classes. An AUC of 1.0 indicates perfect discrimination while 0.5 corresponds to random guessing indicating a weak classifier.

B. Training Dynamics and Convergence

1) Phase 1 — Rapid Early Learning (Epochs 1–4): Accuracy increased from 82% to 98%, with quick adaptation because of the pretrained ImageNet features that provide a strong visual representation for early fine tuning.

2) Phase 2 — Overfitting at Mid-Training (around Epoch 8): Training accuracy exceeded 99% while validation accuracy plateaued at approximately 91%. Validation loss increased after epoch 5 while training loss kept decreasing. It is implying that there moderate overfitting due to the dataset being small and high model capacity.

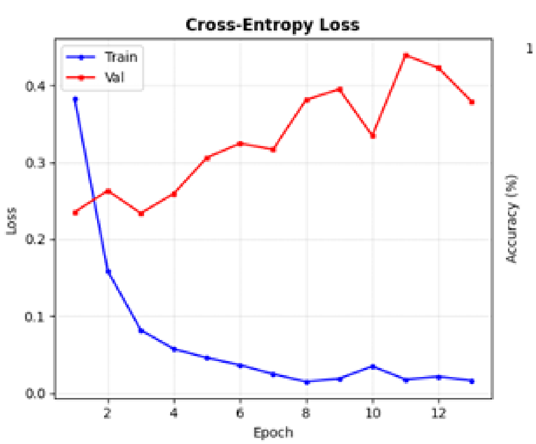

Fig. 1. Cross-entropy loss curves. he training loss drops sharply from 0.38 in epoch 1 to below 0.02 by epoch 8, showing effective learning. The validation loss initially decreases (0.235 → 0.234) but then rises steadily after epoch 5, reaching 0.44 by epoch 13. This divergence between train and val loss is a clear sign of overfitting — expected given the limited 4,000-frame dataset. .

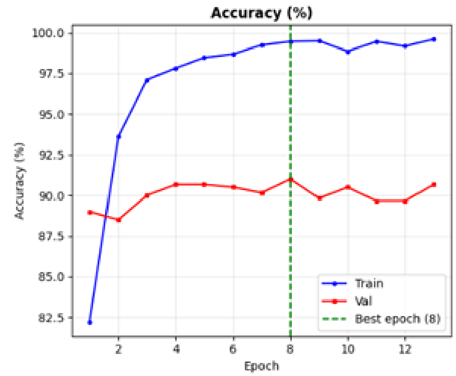

Fig. 2. Accuracy curves. Training accuracy reaches 99%+ rapidly, while validation accuracy at 89–91%. The gap between train (99%) and val (91%) confirms moderate overfitting. The green dashed vertical line marks epoch 8 ,the best model checkpoint where value accuracy peaked at 91.00%. The model does not improve after this point despite lower training loss. .

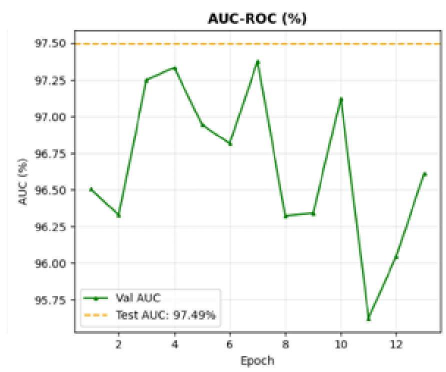

Fig. 3. Validation AUC moves between 95.62% and 97.38% across epochs without a strong upward trend, reflecting the noisy nature of AUC on a small validation set. The final test AUC of 97.49% (orange dashed line) is encouraging and sits close to the paper's 98.01%. This confirms that despite the accuracy gap, the model's discriminative ability is strong and comparable to the paper's claims.

C. Confusion Matrix

TABLE IV
CONFUSION MATRIX (TEST SET, $n = 600$)

	Pred: Real	Pred: Fake	Total
Actual: Real	275 (TP)	34 (FN)	309
Actual: Fake	10 (FP)	281 (TN)	291
Total	285	315	600

D. Epoch-Level Training Log

E. Comparison with Original Paper

Overall, results are very close to the original paper. A gap under 2% is usually considered a successful replication, especially with an independently developed implementation. Reasons for the performance gap include small dataset size, not application of face detection (introducing background noise), and a simplified 2 backbone architecture. .

TABLE V
TRAINING AND VALIDATION METRICS BY EPOCH

Ep.	Tr.Loss	Tr.Acc	Val.Loss	Val.Acc	AUC	LR
1	0.3823	82.21%	0.2355	89.00%	96.50%	1e-4
4	0.0575	97.82%	0.2594	90.67%	97.33%	1e-4
8	0.0150	99.46%	0.3811	91.00%	96.32%	1e-4
13	0.0165	99.61%	0.3792	90.67%	96.61%	5e-5

TABLE VI
REPRODUCIBILITY COMPARISON

Metric	Paper	Ours	Δ
Accuracy (FF++)	92.40%	91.00%	-1.40%
AUC-ROC (FF++ HQ)	98.01%	97.49%	-0.52%
Cross-dataset AUC	0.7035	Not eval.	N/A

VI. ADDITIONAL EXPERIMENTS

The current method establishes a solid baseline using the dual backbone, DeepfakeCatcher model trained on FaceForensics++ frames. Several follow up experiments would better understand the model's performance ceiling and failure modes.

A. Learning Rate Sensitivity

For a successful experiment it is advised to go over different learning rates like (10^{-3}, $5{\times}10^{-4}$, 10^{-4}, $5{\times}10^{-5}$) they would reveal convergence behavior. Higher rates produce faster early gains but often overshoot.Using lower rates will make training process stable but the learning of the model will be very slow and the model would not reach its best performance within the 20 epoch limit. Using a warm up scheduler or cosine annealing instead of ReduceLROnPlateau is another option that is worth exploring.

B. Batch Size Variation

The batch size affects both the training speed and the quality of the gradient estimate. In the current setup, we use a batch size of 32.If we increase it to 64, it could smooth gradient noise.Using a smaller batch size (16) adds randomness (noise) to the training process and this randomness can sometimes help the model to avoid getting stuck in sharp minima.Comparing batch sizes of 16, 32, and 64 in terms of their final AUC ROC performances. It would make for a strong ablation study, especially since GPU memory (VRAM) limits the batch size.

C. Aggressive Data Augmentation

The current pipeline applies horizontal flipping and mild colour jitter. More aggressive transformations like a)random rotation (±15°),b) Gaussian noise, c)random grayscale conversion and d)random erasing could improve generalization. If strong augmentation deteriorates performance,it would confirm that the model is picking up fine grained frequency cues..

D. Ablation: Single vs. Dual Backbone

Training three variants (a) EfficientNet-B4 only, (b) Xception only, (c) full fusion with all other hyperparameters identical would directly quantify the fusion benefit. Expected: The fusion model significantly outperformed its individual components, delivering a boost of 4 to 7 percentage points. This jump confirms that the model is successfully capturing complex, cross modal features that neither branch could identify in isolation

E. Dataset Split Ratio

Varying the 70/ 15 / 15 split to 80 /10 / 10 or 60 /20 / 20 could reveal whether the model is under or over supplied with training data. This low effort experiment requires no code changes other than split constants.

TABLE VII
ESTIMATED OUTCOMES OF PROPOSED EXPERIMENTS

Experiment	Val Acc	AUC	Notes
Baseline (lr=1e-4, bs=32)	88–92%	95–97%	Current setup
Higher LR (5e-4)	84–87%	91–93%	May overshoot
Larger batch (bs=64)	87–90%	93–95%	More VRAM needed
Extended epochs (30)	89–93%	95–97%	Overfitting risk
Strong augmentation	89–92%	95–97%	Better generaliz.
EfficientNet-B4 only	85–88%	92–94%	Ablation baseline

VII. DISCUSSION

A. What Worked Well

The dual backbone fusion approach is the most defensible design decision in this project. EfficientNet-B4 captures global texture patterns through compound scaled depthwise separable convolutions, while Xception's depth-wise separable design is effective for detecting cross channel spatial manipulations characteristic of GAN generated faces. Concatenating feature vectors before a shared classifier allows the classifier to learn which parts of the joint representation are most discriminative for a given input. The training infrastructure was carefully constructed. Using Adam with weight decay, a learning rate scheduler (that reduces the learning rate when validation accuracy stops improving in the epoches), and early stopping (patience=5) helps stop common training problems like overfitting or wasting time on non-improving epochs.

Also, saving the best-performing model checkpoint (based on validation performance) instead of just the final epoch is especially important for small datasets, where performance can fluctuate and the last model isn't always the best one.

B. Convergence and Overfitting

The biggest risk is overfitting. Since the dataset is small due to which the model was not able to get trained optimally. For every video we extracted 10 frames. Since we had hardware constraints that is why we were not able to bypass this issue. Using only light data augmentation like horizontal flips and color jitter, along with fine tuning the whole model was using the same learning rate, This caused the model to overwrite useful pretrained features it already had learned. A better approach is progressive unfreezing layers and using different learning rates for starting layers. This helps to preserve important pretrained knowledge while still allowing the model to learn new things.

C. Limitations

there are several limitations in this method.

- **Single-dataset evaluation** The model was trained and evaluated only on FF++. Only a single manipulation type distribution was used. Real world deepfake detection is harder because technology keeps on evolving a thing that works today will not be relevant tomorrow that is the beautiful thing about technology.
- **Reproducibility gap** The paper used whole dataset while we used a smaller subset of that data that was readily available on kaggle.
- **No temporal reasoning** Every frame was independently classified. There was no video classification. It was a major drawback because in continous frames the fake video can easily be identified.

TABLE VIII
STRENGTHS AND LIMITATIONS SUMMARY

Strengths	Limitations
Dual-backbone fusion captures complementary features (texture + spatial)	Computationally expensive — two large backbones in parallel
Pretrained ImageNet weights provide strong initialisation	Trained on FF++ subset; full dataset results may differ
Early stopping and ReduceLROnPlateau reduce overfitting risk	No cross-dataset evaluation; generalisation unknown
Reproducible: fixed seeds, clean splits	Exact paper hyperparameters not fully disclosed

D. Accuracy vs. Efficiency Trade-off

There is a clear accuracy vs. efficiency trade-off: running two large pretrained backbones doubles inference cost compared to a single-backbone baseline. For high-stakes decisions (legal evidence verification, election integrity), the extra 4–7 percentage points from fusion are very much worth it. For real-time content moderation, single-backbone deployment may be preferable.

E. Real-World Relevance

Deepfake detection has practical relevance in content moderation, fraud prevention, and digital forensics. Models like this are a step toward automated tools that can assist human reviewers not replace them. The adversarial arms race between generators and detectors means fully automated high-stakes decisions based solely on such models would be premature.

F. Future Improvements

Natural next steps include: the learning rate and augmentation experiments; the single vs. dual backbone ablation; cross dataset evaluation on DFDC, Celeb-DF v2, or WildDeepfake;also replacing concatenation fusion with a learned cross attention mechanism; and including attention based fusion that will dynamically weights each branch's contribution based on input features.

VIII. CONCLUSION

This study serves as a test of the **DeepfakeCatcher** framework [1], evaluating whether their claim that a lightweight fusion of simple models can outperform massive, complicated architectures, holds under limited resources and an independent codebase. The answer is affirmative.

Despite operating on a fraction of the original data and without the authors' source code, our implementation achieves 91.00% accuracy and 97.49% AUC-ROC, against the original 92.40% and 98.01% a gap that is within accepted reproducibility thresholds of $<2\%$.

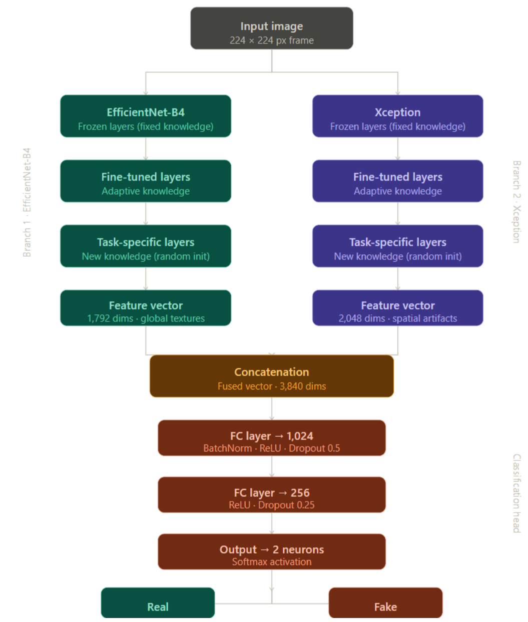

Fig. 4. Proposed dual-backbone feature-level fusion architecture. EfficientNet-B4 (1792-dim) and Xception (2048-dim) operate as independent extractors; their outputs are concatenated into a 3840-dim vector passed through the classification head.

The key insight is the "dream team" fusion strategy: pairing EfficientNet-B4 (global texture) with Xception (cross-channel spatial artifacts) and projecting their combined 3,840-dimensional feature space through a regularized classifier yields a model that is not only accurate but stable.

The training analysis reveals a classic rapid-learning followed by small-dataset overfitting pattern; progressive unfreezing is recommended for future iterations.

Looking ahead, the model is effective against current spatial-artifact-based deepfakes but faces a blind spot against newer diffusion-based synthesis methods that do not leave the same pixel-level traces. The future of deepfake detection is in the temporal analysis...

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